

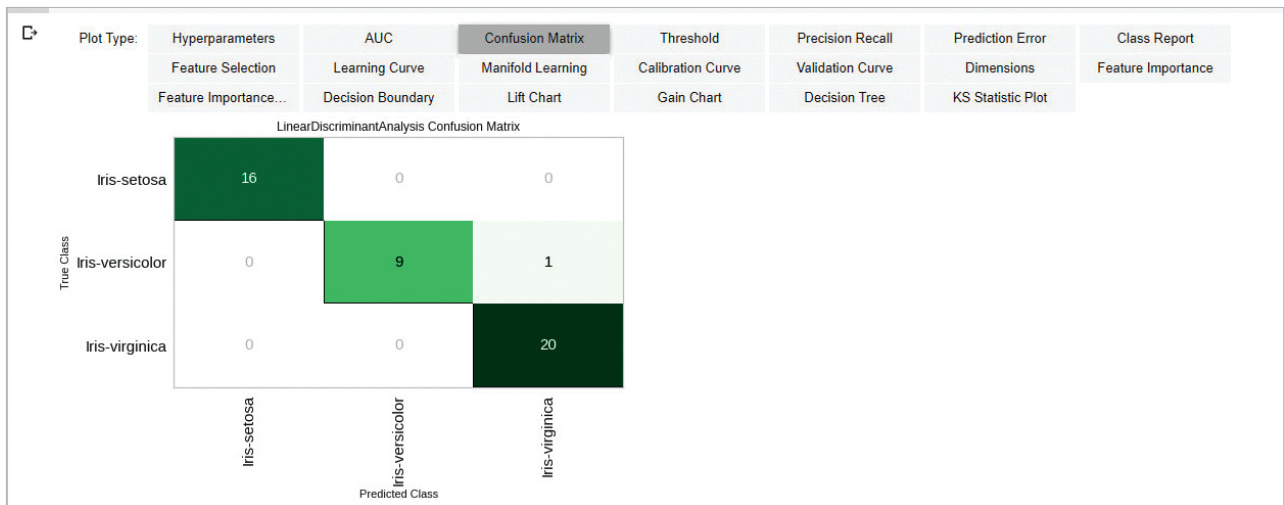


Figure 9: Evaluating and plotting the model performance — confusion matrix

The `predict_model()` function is used for making the predictions of the samples present in the test data.

Step 8: Now plot the model performance, as shown in Figure 9:

```
evaluate_model(tuned_lda)
```

The `evaluate_model()` function is used to develop different performance metrics with minimum effort. You can try them out to see the output. **END** 🐧

References

- [1] <https://pycaret.readthedocs.io/en/latest/index.html>
- [2] <https://pycaret.org/>
- [3] <https://topepo.github.io/caret/>

By: S. Ratan Kumar and Valli Kumari Vatsavayi

S. Ratan Kumar is a passionate user of open source and currently works as an associate professor in the CSE Department of Anil Neerukonda Institute of Technology & Sciences, Visakhapatnam.

Valli Kumari Vatsavayi is a Professor of Computer Science and Systems engineering, Andhra University. Her research interests include Security, Privacy, Machine Learning, and Forensics. She has over 160 research papers of international reputation. She has to her credit several research projects and consultation works.

Continued from page...75

Tips and ideas for learners

Apart from getting a degree or a certification, here are some additional skills that you need to have to become a successful pen tester.

- **Good knowledge of networking:** Understanding computer networks at a deeper level helps pen testers discern vulnerabilities quickly and easily.
- **Systems administration skills:** Understanding and having expert level knowledge of how computer servers work is an integral part of being a tester.
- **Automation skills:** Instead of manually doing a task, learn how to automate a job if it is possible to do so. There are lots of scripting languages available that help

you in automating tasks. Understanding and having a good grasp of such a language will make your path easier towards becoming a successful pen tester.

- **Communication and interpersonal skills:** Soft skills are necessary to

succeed in any work. Pen testers need to pay special attention to communication, as even the most minor things can make the most significant difference if communicated well. **END** 🐧

Resources

Several free and paid resources are available on the Internet that teach penetration testing. Here are some of them that you can check out.

- [1] edx.org: <https://www.edx.org/learn/penetration-testing>
- [2] pentesterlab.com: <https://pentesterlab.com/>
- [3] freecodecamp.org: <https://www.youtube.com/watch?v=3Kq1MifTWCE>
- [4] coursera.org: <https://www.coursera.org/courses?query=pen%20testing>

By: Sufyan bin Uzayr

The author, apart from being a writer, is an educator and entrepreneur. He holds multiple degrees in management, information technology and humanities. He has written seven books so far and is currently the director of a multinational IT company. As an educator, he has taught at universities and institutions worldwide.